Fundamental Theorem of Arithmetic in $\mathbb{Q}$[x]

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Contents

1. Introduction

2. History

Historical background is often nice.

Definition 1. Define **summation**, or $\sum_{i=i_0}^{n} a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\sum_{i=i_0}^{i_0} = a_{i_0} \qquad \sum_{i=i_0}^{n} a_i = a_n + \sum_{i=i_0}^{n-1} a_i$$

Definition 2. Define **product**, or $\prod_{i=i_0}^{n} a_i$ for $n \in \mathbb{Z}^+$ and $n \geq i_0$ as follows:

$$\prod_{i=i_0}^{i_0} = a_{i_0} \qquad \prod_{i=i_0}^{n} a_i = a_n \cdot \prod_{i=i_0}^{n-1} a_i$$

3. Integers

Lemma 1 (NIBZO)**.** *There is no integer strictly between* 0 *and* 1*. Equivalently, if* $n \in \mathbb{Z}$ *and*

$$0 < n < 1,$$

then such an n *cannot exist.*

Proof. Suppose, for contradiction, that there is an integer strictly between 0 and 1. Put

$$S = \{n \in \mathbb{Z} : 0 < n < 1\}.$$

Then S is nonempty and $S \subseteq \mathbb{Z}_{\geq 0}$. By the well-ordering principle, S has a least element. Call it m.

Since $0 < m$, multiplicative closure of positive integers gives $0 < m^2$. Since $m < 1$ and $m > 0$, multiplying the inequality $m < 1$ by the positive integer m gives

$$m^2 < m.$$

Thus m^2 is also an integer satisfying

$$0 < m^2 < m < 1.$$

So $m^2 \in S$, but $m^2 < m$. This contradicts the choice of m as the least element of S. Therefore S is empty. □

4. Rationals

4.1. Construction

Define $\mathbb{Q} \subseteq \mathbb{Z} \times \mathbb{Z}$, where the elements of $\mathbb{Q}$ are of the form (a, b) where $a, b \in \mathbb{Z}, b \neq 0$. Define two operations $+, \cdot$, such that

- $+ : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ where $(a, b) + (c, d) = (ad + bc, bd)$

- $\cdot : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$ where $(a, b) \cdot (c, d) = (ac, bd)$

Also, define an equivalence relation on $\mathbb{Q}$. Two elements $(a, b), (c, d)$ in $\mathbb{Q}$ are equal iff $ad = bc$. Denote equivalence as $(a, b) \sim (c, d)$.

We can also denote (a, b) as $\frac{a}{b}$.

4.2. Ring Axioms

Commutativity of Multiplication:

$$(a, b) \cdot (c, d) = (ac, bd) \qquad \text{and} \qquad (c, d) \cdot (a, b) = (ca, db) = (ac, bd)$$

Commutativity of Addition:

$$(a, b) + (c, d) = (ad + bc, bd) \qquad \text{and} \qquad (c, d) + (a, b) = (cb + ad, db) = (ad + bc, bd)$$

Associativity of Addition:

$$((a, b) + (c, d)) + (e, f) = (a, b) + ((c, d) + (e, f))$$

For the left hand side,

$$(ad + bc, bd) + (e, f) = (adf + bcf + bde, bdf)$$

For the right hand side

$$(a, b) + (cf + de, df) = (adf + bcf + bde, bdf).$$

Associativity of Multiplication:

$$(a, b) \cdot ((c, d) \cdot (e, f)) = ((a, b) \cdot (c, d)) \cdot (e, f)$$

For the left hand side,

$$(a, b) \cdot (ce, df) = (ace, bdf)$$

For the right hand side,

$$(ac, bd) \cdot (e, f) = (ace, bdf).$$

Additive Identity:

$$(a, b) + (0, 1) = (a, b)$$

From the definition, $(a, b) + (0, 1)$ equals

$$(a \cdot 1 + b \cdot 0, b \cdot 1)$$

By the one axiom and zero axiom,

$$(a, b)$$

Multiplicative Identity:

$$(a, b) \cdot (1, 1) = (a, b)$$

From the definition,

$$(a, b) \cdot (1, 1)$$

equals

$$(a \cdot 1, b \cdot 1)$$

By the one axiom,

$$(a, b)$$

Distributivity:

We prove for $x, y, z \in \mathbb{Q}$, we have $x(y + z) = xy + xz$. Suppose $x = (a, b), y = (c, d), z = (e, f)$. Then $x(y+z) = (a, b) \cdot ((c, d) + (e, f)) = (a, b) \cdot (cf + de, df) = (a(cf + de), b(df)) = (acf + ade, bdf)$. We also have $xy = (a, b) \cdot (c, d) = (ac, bd)$ and $xz = (a, b) \cdot (e, f) = (ae, bf)$. We then have $xy + xz = (ac, bd) + (ae, bf) = (acbf + bdae, bdbf) = (acf + dae, dbf) = (acf + ade, bdf)$. Hence $x(y + z) = xy + xz$. Right distributivity follows immediately, since $(y + z)x = x(y + z) = xy + xz = yx + zx$.

Equivalence

$$(a, b) \sim (c, d) \longleftrightarrow ad = bc$$

Symmetry

$$if (a, b) \sim (c, d)$$

then

$$ad = bc$$

and

$$cb = da$$

so

$$(c, d) \sim (a, b)$$

Transitivity

$$(a, b) \sim (c, d), (c, d) \sim (e, f)$$

then

$$ad = bc, cf = de$$

and

$$adcf = bcde$$

and

$$cdaf = cdbe$$

so

$$af = be$$

finally

$$(a, b) \sim (e, f)$$

Well-Defined To show that addition is well defined, we show that $(a, b) + (c, d) = (a, b) + (e, f)$ if $(c, d) \sim (e, f)$. According to our criterion for an equivalence relation, $cf = de$. Therefore, we want to show $(ad + bc, bd) = (af + be, bf)$. To show these are equivalent, we want to show $abdf + b^2ed = abdf + b^2cf$. We cancel $abdf$ and b^2 because b is nonzero to show that $de = cf$, which is true by our equivalence relation.

To show that multiplication is well defined, we show that $(a, b) \cdot (c, d) = (a, b) \cdot (e, f)$ if $(c, d) \sim (e, f)$. According to our criterion for an equivalence relation, $cf = de$. Therefore, we want to show $(ac, bd) = (ae, bf)$. To show these are equivalent, we want to show $acbf = bdae$. Suppose $a = 0$. Then $0 = 0$, which is true. Otherwise, suppose $a \neq 0$. Then because $ab = ac \implies b = c$ when $a \neq 0$, we can cancel ab from both sides to get $cf = de$, which is true by the equivalence $(c, d) = (e, f)$. Hence multiplication is well defined.

Multiplicative Inverse For any (a, b) and $(c, d) \in \mathbb{Q}$, and $a \neq 0$, $\exists (bc, ad)$ where

$$(a, b)(bc, ad) = (abc, abd)$$

and

$$(abc, abd) \sim (c, d)$$

then

$$(a, b) \mid (c, d)$$

Therefore, define $\frac{p}{q}$ for $p, q \in \mathbb{Q}$ and $p \neq 0$ to be the unique solution x to $px = q$. Note also that this is equivalent to the fraction notation of rationals since $\frac{(a,1)}{(b,1)} = (a,b) = \frac{a}{b}$. Then, define p^{-1} to be $\frac{(1,1)}{p}$.

5. The Ring $\mathbb{Q}[x]$

Definition 3 (The polynomial ring $\mathbb{Q}[x]$)**.** We define a polynominal $f(x) \in \mathbb{Q}[x]$ with degree n if it can be written as

$$f(x) = \sum_{0}^{n} a_i x^i$$

where $n \in \mathbb{Z}_{\geq 0}$ and each coefficient a_i lies in $\mathbb{Q}$ and $a_n \neq 0$.

Theorem 2 ($\mathbb{Q}[x]$ is a commutative ring.)**.**

Proof. We check the ring conditions. Addition stays inside $\mathbb{Q}[x]$ because each coefficient $a_i + b_i$ is rational. Multiplication also stays inside $\mathbb{Q}[x]$ because each coefficient c_k is built from finitely many sums and products of rational numbers.

The zero polynomial is the additive identity because adding zero to every coefficient changes nothing. The polynomial $-f(x)$ is the additive inverse of $f(x)$ because each coefficient becomes $a_i + (-a_i) = 0$. The constant polynomial 1 is the multiplicative identity because multiplying by 1 leaves the coefficients unchanged.

The associative and commutative laws come from the same laws in $\mathbb{Q}$, coefficient by coefficient. For instance, addition is associative because

$$(a_k + b_k) + c_k = a_k + (b_k + c_k)$$

for each coefficient. The multiplication laws and distributivity are checked in the same way from the rational-number laws.

Finally, the zero polynomial has constant coefficient 0, while the constant polynomial 1 has constant coefficient 1. Since $0 \neq 1$ in $\mathbb{Q}$, have $0_{\mathbb{Q}[x]} \neq 1_{\mathbb{Q}[x]}$. $\square$

Definition 4. For $f(x) \in \mathbb{Q}$[x], let the **norm**, or $\|f(x)\|$ be equal to

$$\|f(x)\| = 2^{\deg f(x)}$$

with the following properties:

- $\|f(x)\| \cdot \|g(x)\| = \|f(x)g(x)\|$

  Proof. By Multiplied Sums, if $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{j=0}^{m} b_j x^j$, with n and m the degrees of f and g respectively, then

$$f(x)g(x) = \sum_{i=0}^{n}\sum_{j=0}^{m} a_i b_j x^i x^j = \sum_{i=0}^{n}\sum_{j=0}^{m} a_i b_j x^{i+j}$$

Notice that to find the degree of $f(x)g(x)$, we take the maximum value of $i+j$ which is their respective maximums, $n+m$, so:

$$\|f(x)g(x)\| = 2^{n+m} = 2^n 2^m = \|f(x)g(x)\|$$

□

- For any $c(x) \in \mathbb{Q}[\mathrm{x}]^{\times}$, $\|c(x)\| = 1$.

  Proof. Since $\deg c(x) = 0$, $\|c(x)\| = 2^0 = 1$. □

Definition 5 (Zero, one, and negatives)**.** The zero element of $\mathbb{Q}[x]$ is the polynomial $0_{Qx} = 0$. Define the degree of 0_{Qx} to be $-\infty$. The one element of $\mathbb{Q}[x]$ is the constant polynomial

$$1_{\mathbb{Q}[x]} = 1$$

If

$$f(x) = \sum_{0}^{n} a_i x^i$$

then

$$-f(x) = \sum_{0}^{n} -a_i x^i$$

Theorem 3 ($\mathbb{Q}[x]$ has no zero divisors)**.** *If $f(x), g(x) \in \mathbb{Q}[x]$ and*

$$f(x)g(x) = 0,$$

then $f(x) = 0$ or $g(x) = 0$.

Proof. Suppose $f \neq 0$ and $g \neq 0$. Then

$$\deg(fg) = \deg(f) + \deg(g) \geq 0,$$

so fg is nonzero. This contradicts $fg = 0$. Hence at least one of f and g must be zero. □

Corollary 4 (Cancellation)**.** *If $h(x) \neq 0$ and*

$$h(x)f(x) = h(x)g(x),$$

then $f(x) = g(x)$.

Proof. Subtract the right side from the left:

$$h(f - g) = 0.$$

Since $h \neq 0$ and $\mathbb{Q}[x]$ has no zero divisors, we must have $f - g = 0$. Thus $f = g$. □

5.1. Basic Arithmetic

Lemma 5 (Uniqueness of zero and one)**.** *The additive identity and multiplicative identity in $\mathbb{Q}[x]$ are unique. In particular, if $f(x) + z(x) = f(x)$ for every $f(x) \in \mathbb{Q}[x]$, then $z(x) = 0$. Similarly, if $f(x)u(x) = f(x)$ for every $f(x) \in \mathbb{Q}[x]$, then $u(x) = 1$.*

Proof. Suppose z acts like an additive identity for f, so $f + z = f$. Add $-f$ to both sides:

$$(-f) + (f + z) = (-f) + f.$$

By associativity and the definition of additive inverse, this becomes

$$0 + z = 0.$$

Since 0 is the additive identity, $z = 0$.

Now suppose u acts like a multiplicative identity for every polynomial. Test this on the constant polynomial 1:

$$1 \cdot u = 1.$$

But $1 \cdot u = u$. Hence $u = 1$. □

Lemma 6 (Uniqueness of additive inverses)**.** *If $u(x)$ and $v(x)$ are both additive inverses of $f(x)$, then $u(x) = v(x)$.*

Proof. Assume

$$f + u = 0 \qquad \text{and} \qquad f + v = 0.$$

Then $f + u = f + v$. Add $-f$ to both sides:

$$(-f) + (f + u) = (-f) + (f + v).$$

By associativity,

$$((-f) + f) + u = ((-f) + f) + v.$$

Thus

$$0 + u = 0 + v,$$

so $u = v$. □

Lemma 7 (Zero and negatives)**.** *For all $f(x), g(x) \in \mathbb{Q}[x]$,*

$$f(x) \cdot 0 = 0, \qquad -(-f(x)) = f(x), \qquad (-f(x))(-g(x)) = f(x)g(x).$$

Proof. First,

$$f \cdot 0 = f(0 + 0) = f \cdot 0 + f \cdot 0.$$

Adding the additive inverse of $f \cdot 0$ to both sides gives $f \cdot 0 = 0$.

Next, since

$$(-f) + f = 0,$$

the polynomial f is an additive inverse of $-f$. By uniqueness of additive inverses,

$$-(-f) = f.$$

For the product of negatives, first observe that

$$fg + (-f)g = (f + (-f))g = 0 \cdot g = 0.$$

Therefore $(-f)g = -(fg)$. Applying this with g replaced by $-g$ gives

$$(-f)(-g) = -(f(-g)).$$

Also,

$$fg + f(-g) = f(g + (-g)) = f \cdot 0 = 0,$$

so $f(-g) = -(fg)$. Hence

$$(-f)(-g) = -(-(fg)) = fg.$$

□

5.2. Divisibility

Definition 6 (Divisibility)**.** For $f(x), g(x) \in \mathbb{Q}[x]$, we say $f(x)$ divides $g(x)$, written

$$f(x) \mid g(x),$$

if there exists $q(x) \in \mathbb{Q}[x]$ such that

$$g(x) = f(x)q(x).$$

Definition 7 (Greatest Common Divisor)**.** For $f(x), g(x) \in \mathbb{Q}[x]$, we defince $\gcd(f(x), g(x)) = d(x)$ to satisfy:

- $d(x) \mid f(x)$ and $d(x) \mid g(x)$
- $\forall h(x) \in \mathbb{Q}$[x] s.t. $h(x) \mid f(x)$ and $h(x) \mid g(x)$, $\|h(x)\| \leq \|d(x)\|$

6. The Division Algorithm in $\mathbb{Q}[x]$

For $\mathbb{Q}[x]$, the norm plays the role that size plays in integer division. Since norms of nonzero polynomials are positive integers, we can use the well-ordering principle to pick a remainder of smallest norm.

Theorem 8 (Division algorithm in $\mathbb{Q}[x]$)**.** *Let $f(x), g(x) \in \mathbb{Q}[x]$ with $g(x) \neq 0$. Then there exist unique polynomials $q(x), r(x) \in \mathbb{Q}[x]$ such that*

$$f(x) = q(x)g(x) + r(x),$$

where either

$$r(x) = 0$$

or

$$\|r(x)\| < \|g(x)\|.$$

Proof. **Existence.** Look at all possible remainders after subtracting multiples of g from f:

$$S = \{\, f(x) - q(x)g(x) : q(x) \in \mathbb{Q}[x] \,\}.$$

If $0 \in S$, then $f = qg$ for some q, so

$$f = qg + 0,$$

and the result holds with $r = 0$.

Now suppose $0 \notin S$. Then every element of S is nonzero, so each element has a norm in $\mathbb{Z}^+$. Define

$$D = \{\, \|s(x)\| : s(x) \in S \,\} \subseteq \mathbb{Z}^+.$$

The set D is nonempty because $f - 0g = f$ lies in S. By the well-ordering principle, D has a smallest element. Choose $r \in S$ with minimum norm. Since $r \in S$, there exists $q \in \mathbb{Q}[x]$ such that

$$r = f - qg.$$

Equivalently,

$$f = qg + r.$$

It remains to show that $\|r(x)\| < \|g(x)\|$. Suppose instead that

$$\|r(x)\| \geq \|g(x)\|.$$

Write

$$r(x) = a_m x^m + \cdots \qquad \text{and} \qquad g(x) = b_n x^n + \cdots ,$$

where $a_m, b_n \neq 0$, $m = \deg(r)$, and $n = \deg(g)$. Since $\|r(x)\| \geq \|g(x)\|$ means $2^m \geq 2^n$, we get $m \geq n$. Since $b_n \neq 0$ and $b_n \in \mathbb{Q}$, the rational number a_m/b_n exists. Define

$$r^*(x) = r(x) - \frac{a_m}{b_n} x^{m-n} g(x).$$

The second term has leading term $a_m x^m$, so the leading term of r cancels. Thus either $r^* = 0$ or

$$\|r^*(x)\| < \|r(x)\|.$$

Also,

$$\begin{aligned} r^* &= f - qg - \frac{a_m}{b_n} x^{m-n} g \\ &= f - \left(q + \frac{a_m}{b_n} x^{m-n} \right) g. \end{aligned}$$

Therefore $r^* \in S$. If $r^* = 0$, then $0 \in S$, which is not the case we are in. If $r^* \neq 0$, then r^* is in S and has smaller norm than r, contradicting how we chose r. Therefore our assumption was false, and

$$\|r(x)\| < \|g(x)\|.$$

Uniqueness. Suppose

$$f = q_1 g + r_1 = q_2 g + r_2,$$

where r_1 and r_2 are both either zero or have norm less than $\|g(x)\|$. Subtracting gives

$$g(q_1 - q_2) = r_2 - r_1.$$

Assume $q_1 \neq q_2$. Then $q_1 - q_2 \neq 0$, and since $\mathbb{Q}[x]$ has no zero divisors,

$$g(q_1 - q_2) \neq 0.$$

Hence

$$\begin{aligned}\|g(x)(q_1(x) - q_2(x))\| &= 2^{\deg(g(q_1-q_2))}\\ &= 2^{\deg(g)+\deg(q_1-q_2)}\\ &= \|g(x)\| \cdot \|q_1(x) - q_2(x)\|\\ &\geq \|g(x)\|.\end{aligned}$$

But $r_2 - r_1$ is either zero or has norm strictly less than $\|g(x)\|$, since each of r_1 and r_2 is either zero or has norm less than $\|g(x)\|$, so their degrees are both less than $\deg(g)$ when they are nonzero. This is impossible. Therefore $q_1 = q_2$. Substituting back gives $r_1 = r_2$. □

7. Euclid's Algorithm

Definition 8. Define **Euclids Algorithm** on $f(x), g(x) \in \mathbb{Q}$[x] to produce $r, q \in \mathbb{Q}$[x] such that

$$f = q_1 g + r_1 \qquad g = q_2 r_1 + r_2 \qquad r_{n-2} = q_n r_{n-1} + r_n$$

Lemma 9. *For any $f(x), g(x) \in \mathbb{Q}$[x], the Euclidean Algorithm generates an $n \in \mathbb{Z}^+$ such that $r_n = 0$.*

Proof. Let $S = \{f(x) \in \mathbb{Q}[\mathrm{x}] \mid \exists\, g(x) \text{ s.t. } r_i \neq 0\ \forall i \in \mathbb{Z}\}$. For the sake of contradiction, S is not empty. By the well-ordering principle, there exists a least element l. Performing Euclid's Algorithm on (l, g), we have:

$$g = q_1 * l + r_1$$
$$l = q_2 r_1 + r_2$$

If (g, l) does not reach 0, (l, r_1) will also not reach 0, as they generate the same values. Thus, $r_1 \in S$, but $r_l < l$ by the division algorithm, which is a contradiction. Thus, S must be empty. □

Lemma 10. *For any $f(x), g(x) \in \mathbb{Q}$[x], if the Euclidean Algorithm generates $r_n = 0$, $r_{n-1} = \gcd(f, g)$.*

Proof. Let $S = \{f(x) \in \mathbb{Q}[\mathrm{x}] \mid \exists\, g(x) \text{ s.t. } r_{n-1} \neq \gcd(f,g)\}$. For the sake of contradiction, S is not empty. By the well-ordering principle, there exists a least element l. Performing Euclid's Algorithm on (l, g), we have:

$$g = q_1 * l + r_1$$

If $\gcd(g, l) = d$, $d \mid g$ and $d \mid l$. Since $b > l$, $\gcd(l, r_1)$ remains d. However, the Euclidean Algorithm on (r_1, l) generates the same values as (b, l). Thus, $r_1 \in S$, but $r_l < l$ by the division algorithm, which is a contradiction. Thus, S must be empty. □

8. Conclusion

Conclude with the big ideas!

Appendix A: Summation and Product Lemmas

Lemma 11. $\sum\limits_{i=i_0}^{n} a_i + \sum\limits_{i=i_0}^{n} b_i = \sum\limits_{i=i_0}^{n} (a_i + b_i)$

Proof. Note that

$$\sum_{i=i_0}^{i_0} a_i + \sum_{i=i_0}^{i_0} b_i = a_{i_0} + b_{i_0} = \sum_{i=i_0}^{i_0} (a_i + b_i)$$

Let $S = \left\{ k \in \mathbb{Z}^+ \;\middle|\; \sum\limits_{i=i_0}^{i_0+k} a_i + \sum\limits_{i=i_0}^{i_0+k} b_i \neq \sum\limits_{i=i_0}^{i_0+k} (a_i + b_i) \right\}$. For the sake of contradiction, S is not empty.
As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.
If $k = 1$:

$$\sum_{i=i_0}^{i_0+1} a_i + \sum_{i=i_0}^{i_0+1} b_i = a_{i_0+1} + a_{i_0} + b_{i_0+1} + b_{i_0} = (a_{i_0+1} + b_{i_0+1}) + (a_{i_0} + b_{i_0}) = \sum_{i=i_0}^{i_0+1} (a_i + b_i)$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l-1 \in \mathbb{Z}$ yet $l-1 \notin S$,

$$\sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i = \sum_{i=i_0}^{i_0+l-1} (a_i + b_i)$$

$$a_l + b_l + \sum_{i=i_0}^{i_0+l-1} a_i + \sum_{i=i_0}^{i_0+l-1} b_i = a_l + b_l + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i)$$

$$a_l + \sum_{i=i_0}^{i_0+l-1} a_i + b_l + \sum_{i=i_0}^{i_0+l-1} b_i = (a_l + b_l) + \sum_{i=i_0}^{i_0+l-1} (a_i + b_i)$$

$$\sum_{i=i_0}^{i_0+l} a_i + \sum_{i=i_0}^{i_0+l} b_i = \sum_{i=i_0}^{i_0+l}(a_i + b_i)$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Lemma 12. *Sum Constants:* $\sum\limits_{i=i_0}^{n} ca_i = c \sum\limits_{i=i_0}^{n} a_i$

Proof. Note that

$$\sum_{i=i_0}^{i_0} ca_i = ca_{i_0} = c\sum_{i=i_0}^{i_0} a_i$$

and that

$$\sum_{i=i_0}^{i_0+1} ca_i = ca_{i_0} + ca_{i_0+1} = c\sum_{i=i_0}^{i_0+1} a_i$$

Let $S = \left\{k \in \mathbb{Z}^+ \;\middle|\; \sum\limits_{i=i_0}^{i_0+k} ca_i \neq c \sum\limits_{i=i_0}^{i_0+k} a_i\right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the well-ordering principle, there exists a least element $l \in S$. Since $1 \notin S$, $l > 1$ by NIBZO. As $l-1 \in \mathbb{Z}$ yet $l-1 \notin S$,

$$\sum_{i=i_0}^{i_0+l-1} ca_i = c \sum_{i=i_0}^{i_0+l-1} a_i$$

$$ca_l + \sum_{i=i_0}^{i_0+l-1} ca_i = ca_l + c \sum_{i=i_0}^{i_0+l-1} a_i$$

$$\sum_{i=i_0}^{i_0+l} ca_i = c\sum_{i=i_0}^{i_0+l} a_i$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. $\square$

Lemma 13. *Multiplied Sums:* $\sum_{i=i_0}^{n} a_i \cdot \sum_{j=j_0}^{m} b_j = \sum_{i=i_0}^{n} \sum_{j=j_0}^{m} (a_i b_j)$

Proof. By Sum Constants, we know that

$$\sum_{i=i_0}^{n} a_i \cdot \sum_{j=j_0}^{m} b_j = \sum_{i=i_0}^{n} (a_i \cdot \sum_{j=j_0}^{m} b_j)$$
$$\sum_{i=i_0}^{n} a_i \cdot \sum_{j=j_0}^{m} b_j = \sum_{i=i_0}^{n} (\sum_{j=j_0}^{m} a_i b_j)$$

□

Lemma 14. $\prod_{i=i_0}^{n} a_i \prod_{i=i_0}^{n} b_i = \prod_{i=i_0}^{n} a_i b_i$

Proof. Note that

$$\prod_{i=i_0}^{i_0} a_i \prod_{i=i_0}^{i_0} b_i = a_{i_0} b_{i_0} = \prod_{i=i_0}^{i_0} a_i b_i$$

Let $S = \left\{ k \in \mathbb{Z}^+ \,\middle|\, \prod_{i=i_0}^{i_0+k} a_i \prod_{i=i_0}^{i_0+k} b_i \neq \prod_{i=i_0}^{i_0+k} a_i b_i \right\}$. For the sake of contradiction, S is not empty. As $S \subseteq \mathbb{Z}^+$, by the Well-Ordering Principle, there exists a least element $l \in S$.
If $k = 1$:

$$\prod_{i=i_0}^{i_0+1} a_i \prod_{i=i_0}^{i_0+1} b_i = a_{i_0+1} a_{i_0} \cdot b_{i_0+1} b_{i_0} = (a_{i_0+1} b_{i_0+1})(a_{i_0} b_{i_0}) = \prod_{i=i_0}^{i_0+1} a_i b_i$$

Thus, $1 \notin S$ and so $l > 1$ by NIBZO. Since $l - 1 \in \mathbb{Z}$ yet $l - 1 \notin S$,

$$\prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i = \prod_{i=i_0}^{i_0+l-1} a_i b_i$$
$$a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i \prod_{i=i_0}^{i_0+l-1} b_i = a_l b_l \cdot \prod_{i=i_0}^{i_0+l-1} a_i b_i$$
$$(a_l \prod_{i=i_0}^{i_0+l-1} a_i) \cdot (b_l \prod_{i=i_0}^{i_0+l-1} b_i) = (a_l b_l) \prod_{i=i_0}^{i_0+l-1} a_i b_i$$
$$\prod_{i=i_0}^{i_0+l} a_i \prod_{i=i_0}^{i_0+l} b_i = \prod_{i=i_0}^{i_0+l} a_i b_i$$

However, since $l \in S$, this is a contradiction. Thus, S must be empty. □